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Launching Off a Planet

Sophie Chen

[12 pts]

Dr. Wen is visiting a planet of radius $R = 6 \times 10^6$ m and mass $M = 2.16 \times 10^{24}$ kg. An orbit is designed at an altitude of $h = 2.5 \times 10^4$ m above the surface.

Now Dr. Wen is going to leave. Being strong, he decides to propel himself (mass $m = 60$ kg) using a spring of stiffness $k = 1.2 \times 10^7$ N/m. This gives him an initial velocity of magnitude v_0, directed at an angle θ above the local tangent to the ground.

Assumptions

1. The planet is a uniform sphere; only its gravitation acts on Dr. Wen after launch.
2. The launch happens from a small platform at the surface, and the spring's compression is much shorter than the planet's radius, so the change in gravitational potential energy during spring expansion is negligible.
3. For the flat-ground part (Q2), the surface is locally flat and the gravitational field is uniform with the surface value $g = GM/R^2$.

Questions.

Q1. Orbital speed. Dr. Wen enters orbit by himself, maintaining a uniform linear speed while travelling around the circular orbit at altitude h. Find this speed v_{orbit}.

Q2. Vertical launch speed. Now approximate the surface as locally flat, with a uniform vertical gravitational field (take $+x$ tangent to the planet and $+y$ normal, upward). Using kinematics for constant-acceleration vertical motion, find the vertical component of the initial velocity.

Q3. Initial velocity. Using the results of parts (1) and (2), calculate v_0 (and the angle θ).

Q4. Spring compression. When the spring is released, its elastic potential energy converts completely into the traveller's initial kinetic energy. Neglecting the change in gravitational potential energy during spring expansion, find the preloaded compression distance x of the spring.

Q5. Orbital force. The traveller moves steadily along the circular orbit. Calculate the magnitude of the universal gravitational force exerted by the planet on the traveller.